COMPUTER ENGINEERING

QUEZON CITY

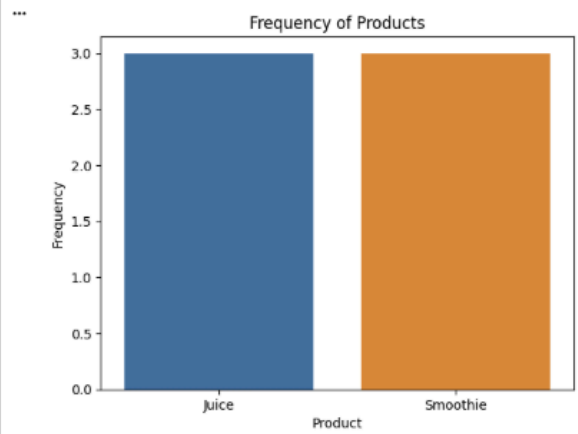
COURSE CODE/TITLE:	CPE 031 - Visualizations and Data Analysis	**SECTION:**	CPE21S1
DATE PERFORMED:	10 - 10 - 2026	**DATE SUBMITTED:**	10 - 10 - 2026
NAME:	Canoy Hail B. Corro Yazinel	**INSTRUCTOR:**	Engr. Jimlord M. Quejado

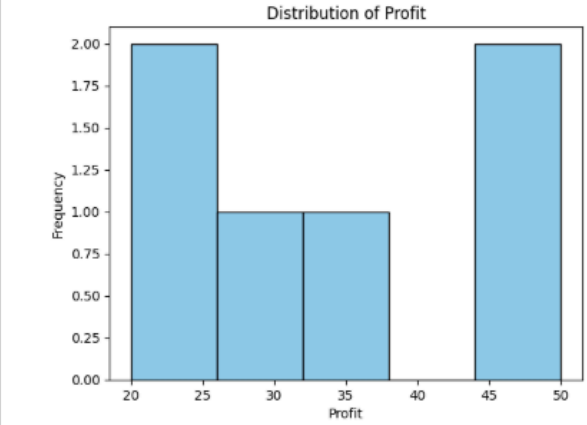
ACTIVITY NO. 9 **Exploring Data Visually**

1. PROCEDURE OUTPUT

Task 1	

Canoy:	Task:
	Create one countplot for Product and one histogram for Profit.
	Compare their patterns and interpret what they mean.
	
	`#Code here`
	
	`# Countplot for the Product`
	`sns.countplot(x="Product", data=df, hue="Product", legend=False)`
	`plt.title("Frequency of Products")`
	`plt.xlabel("Product")`
	`plt.ylabel("Frequency")`
	`plt.show()`
	
	`# Histogram for the Profit`
	`plt.hist(df["Profit"], bins=5, color="skyblue", edgecolor="black")`
	`plt.title("Distribution of Profit")`
	`plt.xlabel("Profit")`
	`plt.ylabel("Frequency")`
	`plt.show()`





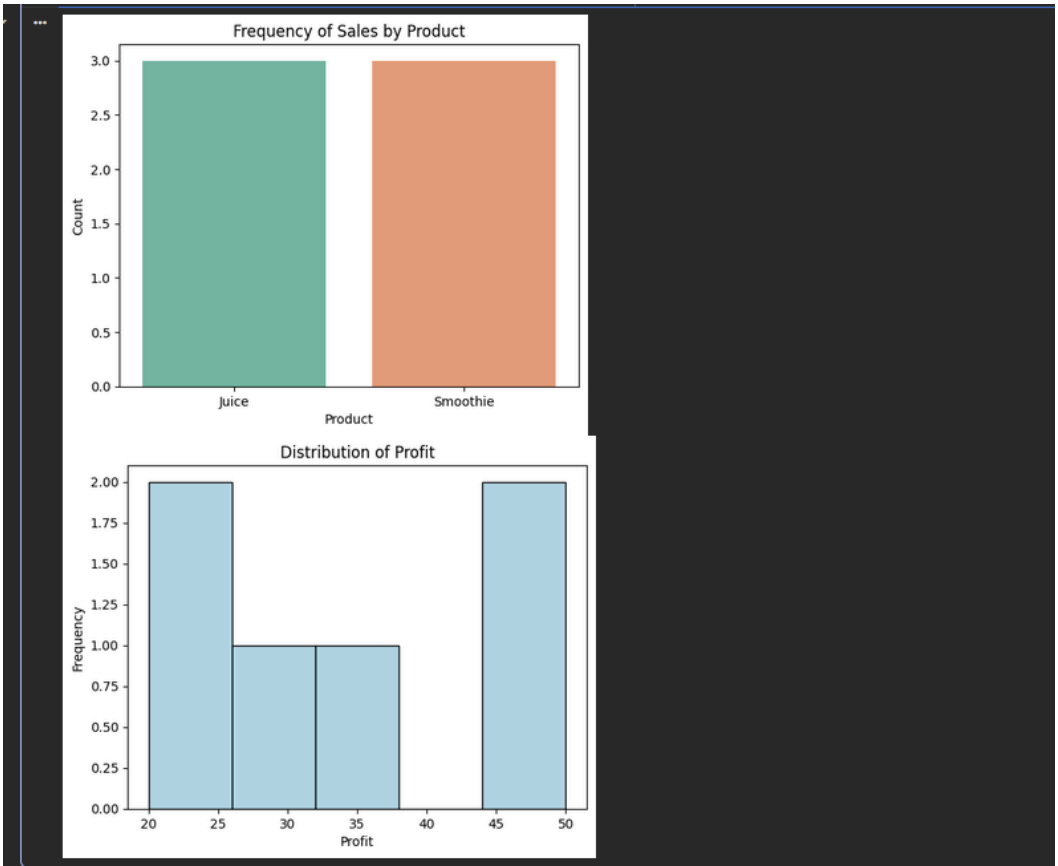
Comparison of their patterns and interpretation

- First the coutplot of the products basically it just shows how often a product appears in the data set while the histogram for the profit shows how the values are distributed. Overall this visualizations help us identify how frequently a product appears and profit happens most.

Corro:

```
# 1. Countplot for Categorical Variable (Product)
sns.countplot(x="Product", data=df, hue="Product", palette="Set2", legend=False)
plt.title("Frequency of Sales by Product")
plt.xlabel("Product")
plt.ylabel("Count")
plt.show()

# 2. Histogram for Quantitative Variable (Profit)
plt.hist(df["Profit"], bins=5, color="lightblue", edgecolor="black")
plt.title("Distribution of Profit")
plt.xlabel("Profit")
plt.ylabel("Frequency")
plt.show()
```



The countplot for Product shows a symmetric distribution, with Juice and Smoothie appearing precisely three times in the data. Meanwhile, the Profit histogram shows a dispersed pattern, with profit values varying between transactions from 20 to 50. Together these graphs suggest that customers purchase Juice and Smoothie at the same rate, but the profit made from each product is different.

Task 2

Canoy:

Task:
Interpret whether higher sales also mean higher profit.
Which region seems to perform best?

*Answer here

- The scatter plot shows a good relationship between sales and profit it shows that sales increase and because of that profit will also increase. The highest sales is smoothie and its from the east region while the lowest sales is juice from the east region.

Corro:

```
# Crosstabulation example
pd.crosstab(df["Region"], df["Product"], values=df["Sales"], aggfunc="mean").fillna(0)
```

Region	Juice	Smoothie
East	90.0	170.0
North	125.0	0.0
South	0.0	150.0
West	0.0	110.0

Task:
Interpret whether higher sales also mean higher profit.
Which region seems to perform best?

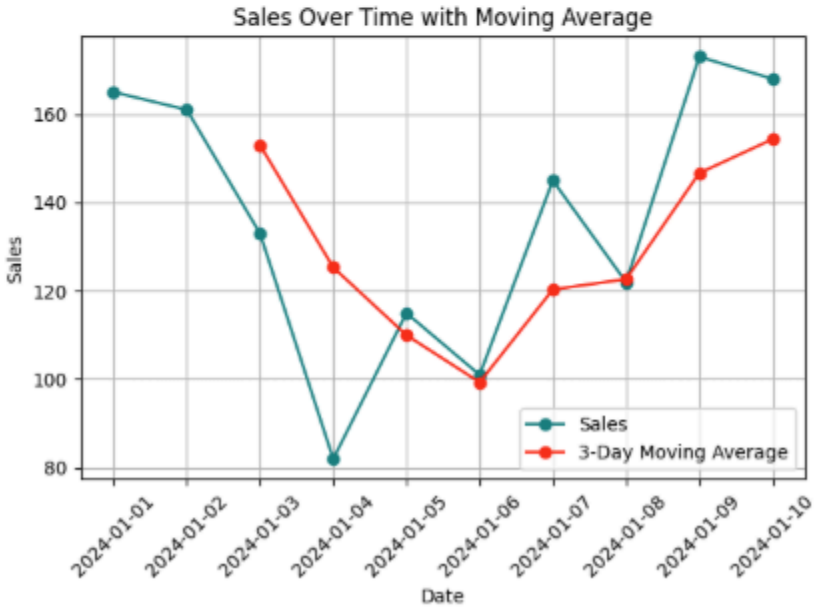
Yes, higher sales directly means higher profit. The scatterplot shows a clear upward trend. Profit increases steadily as sales increase. The best performance is on the East region, it has the highest single sale (170) and highest profit (50) on the graph. The crosstab table shows that the East region is the leader on average Smoothie sales at 170.0.

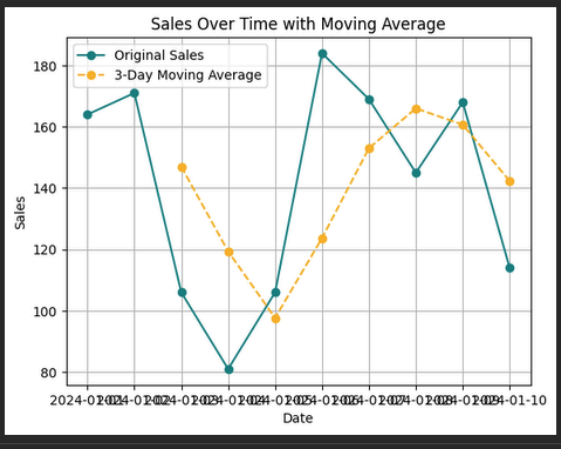
Task 3	
Canoy:	Task: Describe what you observe in the missing data visualization. Which variables need attention before analysis? *Answer here • The missing data from the chart is from th sales column and profit. They both have one missing value while the product and region is complete it still needs attention because the missing values can affect the accuracy of the results.
Corro:	dtype: int64 Task: Describe what you observe in the missing data visualization. Which variables need attention before analysis? In the heatmap of missing data most of the grid is light which means that most of the values are present. There are 2 dark red blocks though, representing missing values. One at row index 2, Profit column, and one at row index 5, Sales column. From the summary table we can see that Sales has 1 missing value and Profit has 1 missing value and Product and Region has 0 missing values. Sales and Profit variables need some cleaning before analysis. These missing values need to be handled by deleting the rows containing those missing values or replacing them with mean values. The chart should be changed by making it blank or coloring it with light color so that it does not cause errors in numerical calculations and visualizations.

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Task 4	

Canoy:	Task:
	Add a moving average line (rolling mean) to smooth fluctuations.
	Hint: Use ts_df["Sales"].rolling(window=3).mean().
	
	`# Generate time-series data`
	`dates = pd.date_range(start="2024-01-01", periods=10)`
	`sales = np.random.randint(80, 180, 10)`
	`ts_df = pd.DataFrame({"Date": dates, "Sales": sales})`
	
	`# To Calculate the 3-day moving average`
	`ts_df["Moving Average"] = ts_df["Sales"].rolling(window=3).mean()`
	
	`# Plot sales and moving average`
	`plt.plot(ts_df["Date"], ts_df["Sales"], marker="o",`
	`color="teal", label="Sales")`
	`plt.plot(ts_df["Date"], ts_df["Moving Average"], marker="o",`
	`color="red", label="3-Day Moving Average")`
	
	`plt.title("Sales Over Time with Moving Average")`
	`plt.xlabel("Date")`
	`plt.ylabel("Sales")`
	`plt.xticks(rotation=45)`
	`plt.legend()`
	`plt.grid(True)`
	`plt.tight_layout()`
	`plt.show()`
	
	 

Corro	Task:
	Add a moving average line (rolling mean) to smooth fluctuations.
	Hint: Use ts_df["Sales"].rolling(window=3).mean().
	
	`# Calculate the 3-day moving average`
	`ts_df["Sales_MA"] = ts_df["Sales"].rolling(window=3).mean()`
	
	`# Plot original time-series`
	`plt.plot(ts_df["Date"], ts_df["Sales"], marker="o", color="teal", label="Original Sales")`
	
	`# Plot moving average line`
	`plt.plot(ts_df["Date"], ts_df["Sales_MA"], marker="o", color="orange", linestyle="--", label="3-Day Moving Average")`
	
	`plt.title("Sales Over Time with Moving Average")`
	`plt.xlabel("Date")`
	`plt.ylabel("Sales")`
	`plt.grid(True)`
	`plt.legend()`
	`plt.show()`
	
	 

Task 5	
Canoy:	Task: Interpret which branch has the highest sales geographically. How can such visualization help in business decision-making? • It shows the different branches based on their locations. The south branch has the higest score hich is 150 while the east is the lowest of 90. Overall this visualizations cn help businesses see which branches sell the highest, middle and lowest.
Corro:	Task: Interpret which branch has the highest sales geographically. How can such visualization help in business decision-making? The South branch is the highest in sales, and is shown as the darkest blue dot at the very bottom of the map. This kind of map lets companies quickly see which sites are making the most money and which are struggling. Managers know where sales are strong and can decide where to spend more advertising money, send more product or open new stores. 5. Supplementary Activity

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

5. Supplementary Activity

Create your own mini exploratory analysis:

1. Choose a dataset (e.g., from Kaggle, or a CSV file you have).
2. Perform:
   - Univariate analysis (1 categorical + 1 quantitative)
   - Bivariate analysis (scatterplot or crosstab)
   - Missing data visualization
   - Time-series or geospatial visualization (choose one)
3. Summarize your key insights using visual interpretation.

Univariate analysis (1 categorical + 1 quantitative)
(Categorical)

```
#Univariate analysis (1 categorical + 1 quantitative)

#for Categoral

#This bar chart is to show ho many student use each AI tools
import pandas as pd
import matplotlib.pyplot as plt

df["AI_Tool_Used"].value_counts().plot(kind="bar")

plt.title("Most Used AI Tools by Students")
plt.xlabel("AI Tool Used")
plt.ylabel("Number of Students")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

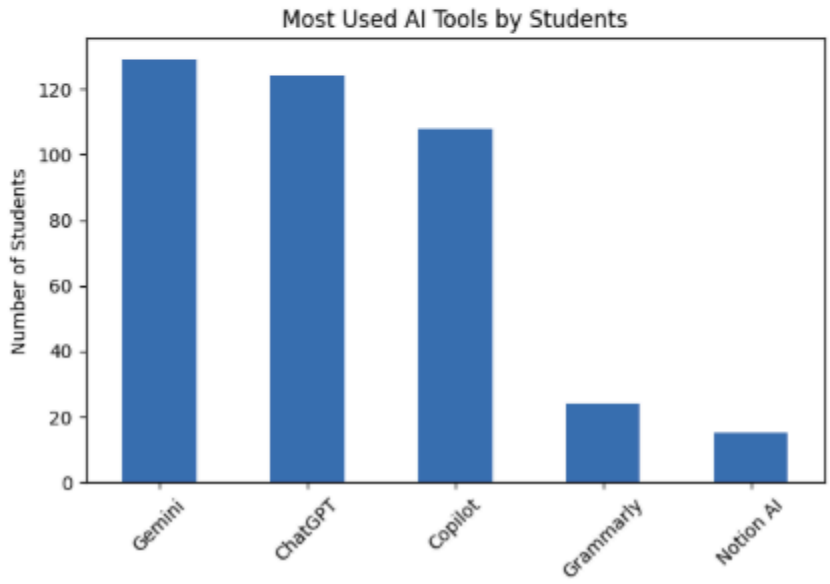


Chart Interpretation: The bar chart basically shows the frequency of AI tools used by students. It shows how much an AI tool is used by the students. Overall this shows us the what is the trend and the most popular AI tools among the students.

Univariate analysis (1 categorical + 1 quantitative)
(Quantitative)

```
#Univariate analysis (1 categorical + 1 quantitative)

#for quantitative

#This bar chart is to see how many usage hours of AI daily
plt.hist(df["Daily_Usage_Hours"].dropna(), bins=10, edgecolor="black")

plt.title("Distribution of Daily AI Usage Hours")
plt.xlabel("Daily Usage Hours")
plt.ylabel("Number of Students")
plt.tight_layout()
plt.show()
```

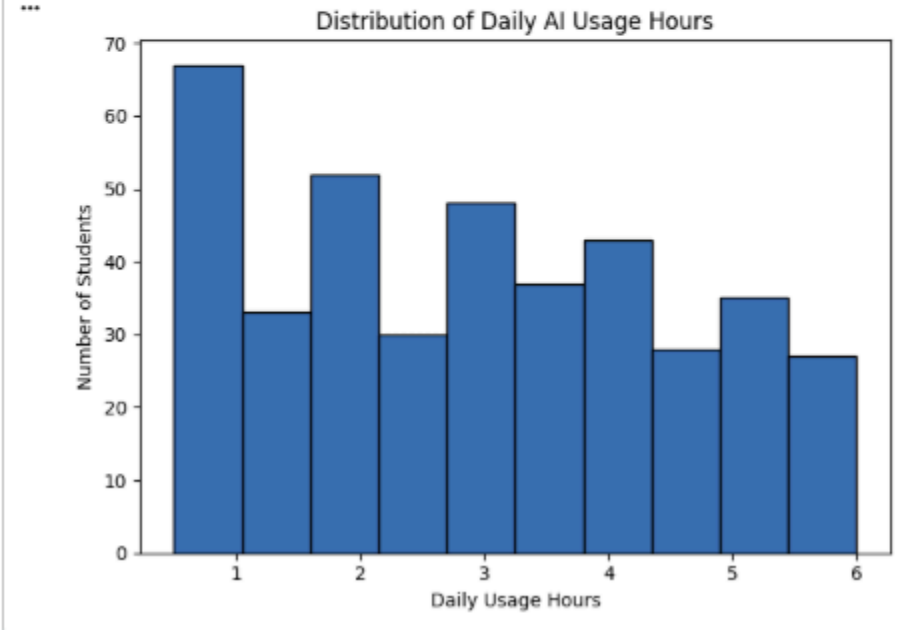


Chart Interpretation: The histogram shows the distributions of daily usage hours among the students. It shows how many daily usage hours each student spends and helps us understand the students daily usage patterns.

Bivariate analysis (scatterplot or crosstab)
(Scatter Plot)

```
#Bivariate analysis (scatterplot or crosstab)

#This scatterplot is used to see the relationship between the daily usage hours
#and its impat on the student grades.

# Convert categories into numeric positions
impact_order = {
    "Negative": 0,
    "No Significant Change": 1,
    "Sight Decline": 2,
    "Slight Decline": 2,
    "Improved": 3
}

df["Impact_Number"] = df["Impact_on_Grades"].map(impact_order)

# Scatterplot
plt.scatter(df["Daily_Usage_Hours"], df["Impact_Number"], alpha=0.6)

plt.title("Daily AI Usage Hours vs. Impact on Grades")
plt.xlabel("Daily Usage Hours")
plt.ylabel("Impact on Grades")
plt.yticks([0, 1, 2, 3], ["Negative", "No Significant Change",
                          "Slight Decline", "Improved"])
plt.tight_layout()
plt.show()
```

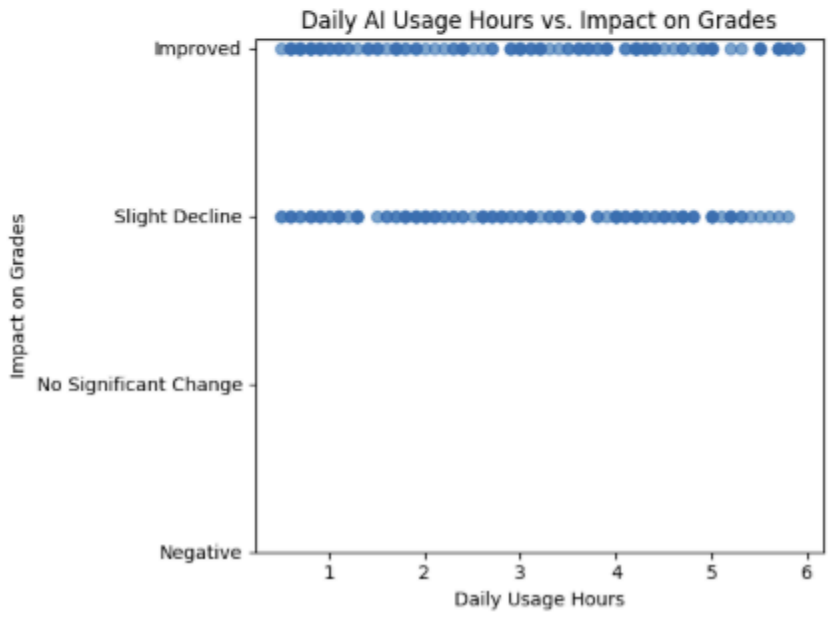


Chart Interpretation: The scatter plot basically shows how the daily use of AI impacts the students grades.

Missing data visualization

```
#Missing data visualization

# for the program to count the missing values in each column
missing_values = df.isnull().sum()

print("Missing Values per Column:")
print(missing_values)

# To keep only columns with missing values
missing_values = missing_values[missing_values > 0]

# Visualize missing values
if not missing_values.empty:
    missing_values.plot(kind="bar", edgecolor="black")

    plt.title("Missing Data Visualization")
    plt.xlabel("Columns")
    plt.ylabel("Number of Missing Values")
    plt.xticks(rotation=45)
    plt.tight_layout()
    plt.show()
else:
    print("No missing values found in the dataset.")
```

```
Missing Values per Column:
Student_ID              0
Age                     0
Gender                  0
Education_Level         0
City                    0
AI_Tool_Used            0
Daily_Usage_Hours       0
Purpose                 0
Impact_on_Grades        0
Satisfaction_Level      0
Impact_Number         131
dtype: int64
```



Chart Interpretation: The chart basically shows how many missing values are in the numbers and as you can see here there are no missing values.

Time-series or geospatial visualization (choose one)
(Geospatial Visualization)

```
#Time-series or geospatial visualization (choose one)

# Calculate average daily AI usage for each city
city_usage = df.groupby("City")["Daily_Usage_Hours"].mean()

# Create bar chart
city_usage.sort_values(ascending=False).plot(
    kind="bar",
    edgecolor="black",
    figsize=(10, 5)
)

plt.title("Average Daily AI Usage by City")
plt.xlabel("City")
plt.ylabel("Average Daily Usage Hours")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

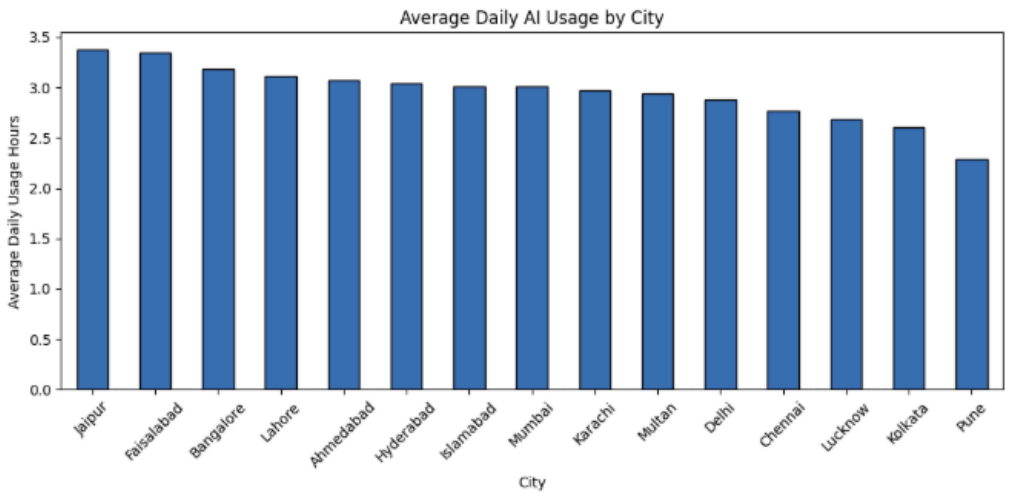


Chart Interpretation: The bar chart compares the average daily AI usage hours across the different cities.

Conclusion (Canoy, Hail B.)
- In this activity I learned more in organizing, making the data look cleaner. This activity really taught me how to analyze a graph more, how to find missing values in a data set, the relationship of two variables in a data sets and many more. Overall this activity helped me understand and learn how to analyze data, find patterns, and present it more clearly.
Conclusion (Corro, Yazinel)
- Through this activity, I learned how to process, clean, and visualize data using tools like pandas, seaborn, and matplotlib. By creating countplots, histograms, and scatterplots, I observed how product sales are evenly distributed, how profit varies across transactions, and how higher sales directly drive higher profits. I also learned how to identify missing values using heatmaps and clean data before running analysis.

Collab Links	
Canoy	Hands_On_Activity_9___Exploring_Data_Visually-1.ipynb
Corro	CORRO-HOA 9: Exploring Data Visually

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **ChatGpt**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Used as a guide in complicated concepts.**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **It was only used as a guide.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the

permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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